(a) Annuity due
(b) An immediate annuity
(c) perpetual annuity
(d) none of these

14. The present value of the perpetual annuity of ₹ 2000 paid monthly at 10 % compound interest is
(a) ₹ 2,40,000 (b) ₹ 6,00,000
(c) ₹ 20,40,000 (d) ₹ 2,00,400

15. Example of contingent annuity is
(a) Installments of payment for a plot of land
(b) An endowment fund to give scholarships to a student
(c) Personal loan from a bank
(d) All the above

Miscellaneous Problems

1. Find the amount of annuity of ₹2000 payable at the end of each year for 4 years of money is worth 10% compounded annually.
$[(1.1)^4 = 1.4641]$

2. An equipment is purchased on an installment basis such that ₹5000 on the signing of the contract and four yearly installments of ₹3000 each payable at the end of first, second, third and the fourth year. If the interest is charged at 5% p.a find the cash down price. $[(1.05)^{-4} = 0.8227]$

3. (i) Find the amount of an ordinary annuity of ₹500 payable at the end of each year for 7 years at 7% per year compounded annually.
$[(1.07)^7 = 1.6058]$

(ii) Calculate the amount of an ordinary annuity of ₹10,000 payable at the end of each half-year for 5 years at 10% per year compounded half-yearly. $[(1.05)^{10} = 1.6289]$

(iii) Find the amount of an ordinary annuity of ₹600 is made at the end of every quarter for 10 years at the rate of 4% per year compounded quarterly. $[(1.01)^{40} = 1.4889]$

(iv) Find the amount of an annuity of ₹2000 payable at the end of every month for 5 years if money is worth 6% per annum compounded monthly. $[(1.005)^{60} = 1.3489]$

4. Naveen deposits ₹250 at the end of each month in an account that pays an interest of 6% per annum compounded monthly, how many months will be required for the deposit to amount to atleast ₹6390?
$[\log(1.1278) = 0.0523, \log(1.005) = 0.0022]$

5. A cash prize of ₹1,500 is given to the student standing first in examination of Business Mathematics by a person every year. Find out the sum that the person has to deposit to meet this expense. Rate of interest is 12% p.a

6. Machine A costs ₹15,000 and machine B costs ₹20,000. The annual income from A and B are ₹4,000 and ₹7,000 respectively. Machine A has a life of 4 years and B has a life of 7 years. Find which machine may be purchased. (Assume discount rate 8% p.a)
$[(1.08)^{-4} = 0.7350, (1.08)^{-7} = 0.5835]$

7. Vijay wants to invest ₹27,000 in buying shares. The shares of the following companies are available to him. ₹100 shares of company *A* at par value; ₹100 shares of company B at a premium of ₹25; ₹100 shares of company *C* at a discount of ₹10; ₹50 shares of company *D* at a premium of 20%. Find how many

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